

HOW MANY SHAFTS DOES IT TAKE TO WEAVE A FACE

Draft jacquard: 242 shafts, 240 pattern blocks. Taquete (weft-faced compound tabby).

Julia Jackson, photographed by Julia Margaret Cameron in 1867 (The Met, public domain), reduced to two wefts on 242 shafts.

Structure. Warp ends alternate pattern, binding, pattern, binding. Binding ends go on shafts 1 and 2 alternately. Each pattern end belongs to one block; block b is shaft $b+2$. Every pick lifts one binding shaft (1, 2, 1, 2 ...) plus the pattern blocks that must hide THIS weft. Two wefts per pass, white and black, and the order alternates each pass (W K, K W) so the binding points checker.

Counts. 480 ends (240 pattern + 240 binding), 1200 picks = 600 passes = 300 pattern rows woven two passes each. Image grid 240 x 300.

Sett. Weft-faced: the warp should not show. Something like 10/2 cotton or linen warp at 20 to 24 epi (8 to 10 per cm), wefts noticeably thicker than the warp (a 3/2 or 5/2 cotton, or two strands), beaten hard so each pass covers about as much length as two ends do width. Check the first 40 passes: a 20-row test square should come out square. If it is short, beat lighter or use a fatter weft.

Floats. By construction no weft floats over more than three ends and no warp end floats over more than two picks, so the cloth is sound at any block count.

The back is the negative. The weft that hides on the face shows on the back.

Files: julia_jacquard.wif (this draft, liftplan form, loads in Fiberworks, WeavePoint, PixeLoom, etc.). Threading strips follow; the liftplan is drawn on the last pages as a reference, read it from the WIF for weaving.



Jacquard: 240 independent pattern ends. There is no paper threading for a TC2; load the WIF or the pixel image (jacquard_pix.png) into the loom software.